

Status Finished**Started** Sunday, 12 January 2025, 4:56 PM**Completed** Sunday, 12 January 2025, 5:14 PM**Duration** 18 mins 32 secs**Grade** 26 out of 40 (65%)**Question 1**

Correct

Mark 1 out of 1

Atbash is an example of a _____

- ☒ A. Substitution Cipher ✓
- ☐ B. Transposition Cipher
- ☐ C. All of the above
- ☐ D. None of the above

The correct answer is:

Substitution Cipher

Question 2

Correct

Mark 1 out of 1

If different keys generate the same ciphertext for the same message, what is this called?

- ☒ A. Collision ✓
- ☐ B. Secure hashing
- ☐ C. MAC
- ☐ D. Key clustering

The correct answer is:

Collision

Question 3

Correct

Mark 1 out of 1

What does DES stand for?

- ☐ A. Data Encryption System
- ☒ B. Data Encryption Standard ✓
- ☐ C. Data Encoding Standard
- ☐ D. Data Encryption Signature

The correct answer is:

Data Encryption Standard

Question 4

Correct

Mark 1 out of 1

Which of the following is not a property or characteristic of a one-way hash function?

- ☐ A. It converts a message of arbitrary length into a value of fixed length.
- ☐ B. Given the digest value, it should be computationally infeasible to find the corresponding message.
- ☐ C. It should be impossible or rare to derive the same digest from two different messages.
- ☒ D. It converts a message of fixed length to an arbitrary length value. ✓

The correct answer is:

It converts a message of fixed length to an arbitrary length value.

Question 5

Correct

Mark 1 out of 1

What is the goal of cryptanalysis?

- ☒ A. To determine the strength of an algorithm ✓
- ☐ B. To increase the substitution function in a cryptographic algorithm.
- ☐ C. To decrease the transposition function in a cryptographic algorithm.
- ☐ D. To determine the permutations used.

The correct answer is:

To determine the strength of an algorithm

Question 6

Correct

Mark 1 out of 1

How many bits make up the effective length of the DES key?

- ☒ A. 56 ✓
- ☐ B. 64
- ☐ C. 32
- ☐ D. 16

The correct answer is:

56

Question 7

Incorrect

Mark 0 out of 1

Which of the following uses a symmetric key and a hashing algorithm?

- ☒ A. HMAC ✗
- ☐ B. Triple-DES
- ☐ C. ISAKMP-OAKLEY
- ☐ D. RSA

The correct answer is:

RSA

Question 8

Correct

Mark 1 out of 1

DES performs how many rounds of permutation and substitution?

- ☒ A. 16 ✓
- ☐ B. 32
- ☐ C. 64
- ☐ D. 56

The correct answer is:

16

Question 9

Correct

Mark 1 out of 1

Which of the following best describes a digital signature?

- ☐ A. A method of transferring a handwritten signature to an electronic document.
- ☐ B. A method to encrypt confidential information.
- ☐ C. A method to provide an electronic signature and encryption.
- ☒ D. A method to let the receiver of the message prove the source and integrity of a message. ✓

The correct answer is:

A method to let the receiver of the message prove the source and integrity of a message.

Question 10

Incorrect

Mark 0 out of 1

What is used to create a digital signature?

- ☐ A. The receiver's private key.
- ☐ B. The sender's public key.
- ☐ C. The sender's private key.
- ☒ D. The receiver's public key. 

The correct answer is:

The sender's private key.

Question 11

Correct

Mark 1 out of 1

Why would a certificate authority revoke a certificate?

- ☐ A. If the user's public key has become compromised
- ☐ B. If the user changed over to using the PEM model that uses a web of trust.
- ☒ C. If the user's private key has become compromised. 
- ☐ D. If the user moved to a new location

The correct answer is:

If the user's private key has become compromised.

Question 12

Correct

Mark 1 out of 1

_____ Cipher replaces bits, characters, or blocks of characters with different bits, characters, or blocks.

- ☒ A. Substitution ✓
- ☐ B. XORing
- ☐ C. Transposition
- ☐ D. Transportation

The correct answer is:

Substitution

Question 13

Incorrect

Mark 0 out of 1

Science of studying and breaking the secrecy of encryption processes, compromising authentication schemes, and reverse-engineering algorithms and keys.

- ☒ A. Cryptology ✗
- ☐ B. Cryptanalysis
- ☐ C. Encryption
- ☐ D. Cryptosystem

The correct answer is:

Cryptanalysis

Question 14

Incorrect

Mark 0 out of 1

_____ refers to the set of all possible keys that can be used to initialize it.

- ☐ A. Cryptology
- ☐ B. Cryptanalysis
- ☒ C. Key Space 
- ☐ D. Encryption algorithm

The correct answer is:

Cryptanalysis

Question 15

Correct

Mark 1 out of 1

20-bit key would have a key space of _____

- ☐ A. 1084572
- ☐ B. 1048578
- ☐ C. 1084576
- ☒ D. 1048576 

The correct answer is:

1048576

Question 16

Correct

Mark 1 out of 1

The frequency of brute force attacks has increased because:

- ☒ A. The use of permutations and transpositions in algorithms has increased. ✓
- ☐ B. As algorithms get stronger, they get less complex, and thus more susceptible to attacks.
- ☐ C. Processor speed and power has increased.
- ☐ D. Key length reduces over time.

The correct answer is:

The use of permutations and transpositions in algorithms has increased.

Question 17

Correct

Mark 1 out of 1

What is the primary purpose of using one-way hashing on user passwords?

- ☐ A. It minimizes the amount of primary and secondary storage needed to store passwords.
- ☒ B. It prevents anyone from reading passwords in plaintext. ✓
- ☐ C. It avoids excessive processing required by an asymmetric algorithm.
- ☐ D. It prevents replay attacks.

The correct answer is:

It prevents anyone from reading passwords in plaintext.

Question 18

Correct

Mark 1 out of 1

What is the definition of an algorithm's work factor?

- ☐ A. The time it takes to encrypt and decrypt the same plaintext.
- ☒ B. The time it takes to break the encryption ✓
- ☐ C. The time it takes to implement 16 rounds of computation.
- ☐ D. The time it takes to apply substitution functions

The correct answer is:

The time it takes to break the encryption

Question 19

Correct

Mark 1 out of 1

Which of the following is a U.S. federal government algorithm developed for creating secure message digests?

- ☐ A. Data Encryption Algorithm
- ☒ B. Digital Signature Standard ✓
- ☐ C. Secure Hash Algorithm
- ☐ D. Data Signature Algorithm

The correct answer is:

Digital Signature Standard

Question 20

Correct

Mark 1 out of 1

What does DEA stand for?

- ☐ A. Data Encoding Algorithm
- ☐ B. Data Encoding Application
- ☒ C. Data Encryption Algorithm ✓
- ☐ D. Digital Encryption Algorithm

The correct answer is:

Data Encryption Algorithm

Question 21

Incorrect

Mark 0 out of 1

When a file is deleted

- ☐ A. The file remains intact.
- ☒ B. The FAT entry for the file is zeroed out so it shows that the area is available for use by a new file. ✗
- ☐ C. The first character of the directory entry file name is changed to a special character.
- ☐ D. All of the above.

The correct answer is:

All of the above.

Question 22

Correct

Mark 1 out of 1

_____ devices prevent altering data on drives attached to the suspect computer and also offer very fast acquisition speeds.

- ☐ A. Encryption
- ☐ B. Imaging
- ☒ C. Write Blocking ✓
- ☐ D. Hashing

The correct answer is:

Write Blocking

Question 23

Incorrect

Mark 0 out of 1

Which duplication method produces an exact replica of the original drive?

- ☐ A. Bit-Stream Copy
- ☒ B. Image Copy ✗
- ☐ C. Mirror Copy
- ☐ D. Drive Image

The correct answer is:

Bit-Stream Copy

Question 24

Correct

Mark 1 out of 1

To verify the original drive with the forensic copy, you use _____.

- ☐ A. a password
- ☒ B. a hash analysis ✓
- ☐ C. disk to disk verification
- ☐ D. none of the above

The correct answer is:

a hash analysis

Question 25

Correct

Mark 1 out of 1

The Windows operating system uses a file name's _____ to associate files with the proper applications.

- ☐ A. Signature
- ☒ B. Extension ✓
- ☐ C. MD5 hash value
- ☐ D. Metadata

The correct answer is:
Extension

Question 26

Incorrect

Mark 0 out of 1

As a good forensic practice, why would it be a good idea to wipe a forensic drive before using it?

- ☒ A. Chain of Custody ✗
- ☐ B. No need to wipe
- ☐ C. Different file and operating systems
- ☐ D. Cross-contamination

The correct answer is:
Cross-contamination

Question 27

Correct

Mark 1 out of 1

The ability to hide data in another file is called

- ☐ A. Encryption.
- ☒ B. Steganography. ✓
- ☐ C. Data parsing.
- ☐ D. A and B.

The correct answer is:
Steganography.

Question 28

Incorrect

Mark 0 out of 1

When two hard drives are on the same data cable, both drives must have which two settings for them to work?

- ☐ A. Default and Cable Select
- ☒ B. Primary and Secondary ❌
- ☐ C. Master and Slave
- ☐ D. First and Second

The correct answer is:

Master and Slave

Question 29

Correct

Mark 1 out of 1

USB drives use _____.

- ☐ A. RAM memory
- ☐ B. Cache memory
- ☒ C. Flash memory ✔️
- ☐ D. None of the above

The correct answer is:

Flash memory

Question 30

Incorrect

Mark 0 out of 1

Which of the following is a proper search technique?

- ☐ A. Manual Browsing
- ☐ B. Keyword Search
- ☐ C. Regular Expression Search
- ☒ D. All of the above ❌

The correct answer is:

Regular Expression Search

Question 31

Correct

Mark 1 out of 1

A file header is which of the following?

- ☒ A. A unique set of characters at the beginning of a file that identifies the file type ✓
- ☐ B. A unique set of characters following the file name that identifies the file type
- ☐ C. A 128-bit value that is unique to a specific file based on its data
- ☐ D. Synonymous with the file extension

The correct answer is:

A unique set of characters at the beginning of a file that identifies the file type

Question 32

Correct

Mark 1 out of 1

Which of the following is not a property of computer evidence?

- ☐ A. Authentic and Accurate.
- ☐ B. Complete and Convincing.
- ☐ C. Duplicated and Preserved.
- ☒ D. Conform and Human Readable. ✓

The correct answer is:

Conform and Human Readable.

Question 33

Correct

Mark 1 out of 1

Which of the following is not a true operating system?

- ☒ A. DOS ✓
- ☐ B. Windows 3.1
- ☐ C. Windows 2000
- ☐ D. UNIX

The correct answer is:

DOS

Question 34

Correct

Mark 1 out of 1

You can use _____, a powerful search tool, to perform keyword searches in Linux and in EnCase software.

- ☒ A. grep ✓
- ☐ B. grub.
- ☐ C. gcc.
- ☐ D. gnu.

The correct answer is:

grep

Question 35

Incorrect

Mark 0 out of 1

You are a computer forensic examiner at a scene and have determined you will seize a Linux server, which according to your source of information contains the database records for the company under investigation for fraud. The best practice for “taking down” the server for collection is to photograph the screen, note any running programs or messages and so on, and

- ☐ A. Use the normal shutdown procedure
- ☒ B. Pull the plug from the wall ✗
- ☐ C. Pull the plug from the rear of the computer
- ☐ D. Ask the user at the scene to shut down the server

The correct answer is:

Use the normal shutdown procedure

Question 36

Incorrect

Mark 0 out of 1

When a forensic copy is made, in what format are the contents of the hard drive stored?

- ☐ A. As compressed images.
- ☐ B. As bootable files.
- ☐ C. As executable files.
- ☒ D. As operating system files. 

The correct answer is:

As compressed images.

Question 37

Incorrect

Mark 0 out of 1

Which of the following is not a type of volatile evidence?

- ☐ A. Routing Tables
- ☐ B. Main Memory
- ☒ C. Log files 
- ☐ D. Cached Data

The correct answer is:

Routing Tables

Question 38

Correct

Mark 1 out of 1

In establishing what evidence is admissible, many rules of evidence concentrate first on the _____ of the offered evidence.

- ☐ A. Relevancy
- ☐ B. Search and Seizure
- ☐ C. Material
- ☒ D. Admissibility 

The correct answer is:

Admissibility

Question 39

Incorrect

Mark 0 out of 1

Which of the following is a proper acquisition technique?

- ☒ A. Disk to Image 
- ☐ B. Disk to Disk
- ☐ C. Sparse Acquisition
- ☐ D. All of the above

The correct answer is:

All of the above

Question 40

Incorrect

Mark 0 out of 1

Traditional crimes that became easier or more widespread because of telecommunication networks and powerful PCs include all of the following except

- ☐ A. Money laundering
- ☐ B. Illegal drug distribution
- ☒ C. DoS attacks 
- ☐ D. Child pornography

The correct answer is:

Illegal drug distribution